

Plant Disease Detection — SVM Classifier

A complete machine learning pipeline using Support Vector Machines to detect plant leaf diseases from sensor/measurement data.

Project Structure

```
plant-disease-svm/
├─ plant_disease_svm.py    # Main script (run this)
├─ requirements.txt        # Python dependencies
├─ output/                 # Generated plots, models, and data
│   ├── plant_disease_data.csv
│   ├── best_svm_model.pkl
│   ├── scaler.pkl
│   └─ *.png plots
└─ README.md
```

Quick Start in VS Code

1. Create Project

```
mkdir plant-disease-svm
cd plant-disease-svm
```

2. Virtual Environment

```
# Windows
python -m venv venv
venv\Scripts\activate

# macOS / Linux
python3 -m venv venv
source venv/bin/activate
```

3. Install Dependencies

```
pip install -r requirements.txt
```

4. Run

```
python plant_disease_svm.py
```

Pipeline Steps

Step	Action	Key Output
1	Generate plant disease dataset	plant_disease_data.csv
2	EDA —feature distributions	Histogram plots
3	Train-test split (75/25)	600 train / 200 test
4	StandardScaler normalization	scaler.pkl

Table 1 – continued

Step	Action	Key Output
5	Train Linear & RBF SVM	Model objects
6	Evaluate —accuracy, precision, recall	Confusion matrices
7	ROC curve & AUC score	<code>roc_curve.png</code>
8	GridSearchCV —tune C, gamma, kernel	<code>best_svm_model.pkl</code>
9	Feature importance analysis	<code>feature_importance.png</code>
10	Predict on 5 new plant samples	Console output

Dataset Features

Feature	Healthy Range	Diseased Range	What It Means
leaf_color_green	150-220	50-140	Higher = healthier
leaf_color_yellow	20-60	90-190	Higher = chlorosis/ disease
spot_size_mm	0-5	5-30	Fungal/bacterial spot size
leaf_texture_roughness	0.0-0.4	0.4-1.0	Surface deformation
humidity_percent	40-70	70-100	Fungal growth conditions
temperature_c	18-28	28-42	Heat stress threshold
soil_ph	6.0-7.5	3.5-5.5	Acidic stress
nitrogen_ppm	90-160	20-80	Nutrient deficiency

SVM Concepts Used

- **Linear kernel:** Straight boundary, interpretable weights
- **RBF kernel:** Non-linear boundary for complex patterns
- **C parameter:** Regularization strength
- **Gamma:** Influence radius of support vectors
- **Support Vectors:** Critical data points defining the margin

VS Code Tips

- Use `# %%` to create Jupyter-like cells
- `Shift + Enter` to run cells interactively
- Install Python extension for IntelliSense
- Set breakpoints and press `F5` to debug

Expected Results

- **Accuracy:** ~95-100% (dataset has clear separability)
- **AUC:** ~1.0 (perfect discrimination)
- **Top disease indicators:** Leaf yellowing, texture roughness, low green color
- **Top health protectors:** High green color, balanced pH, adequate nitrogen